

MRI brain tumor detection and classification using parallel deep convolutional neural networks

Takowa Rahman, Md Saiful Islam *

Department of Electronics and Telecommunication Engineering, Chittagong University of Engineering and Technology, Chattogram, Bangladesh

ARTICLE INFO

Keywords:

Brain tumor detection
Data augmentation
Parallel convolutional neural network
ReLU activation function
Softmax function

ABSTRACT

Convolutional neural network (CNN) is widely used to classify brain tumors with high accuracy. Since CNN collects features randomly without knowing the local and global features and causes overfitting problems, this research proposes a novel parallel deep convolutional neural network (PDCNN) topology to extract both global and local features from the two parallel stages and deal with the over-fitting problem by utilizing dropout regularizer alongside batch normalization. To begin, input images are resized and grayscale transformation is conducted, which helps to reduce complexity. After that, data augmentation has been used to maximize the number of datasets. The benefits of parallel pathways are provided by combining two simultaneous deep convolutional neural networks with two different window sizes, allowing this model to learn local and global information. Three forms of MRI datasets are used to determine the effectiveness of the proposed method. The binary tumor identification dataset-I, Figshare dataset-II, and Multiclass Kaggle dataset-III provide accuracy of 97.33%, 97.60%, and 98.12%, respectively. The proposed structure is not only accurate but also efficient, as the proposed method extracts both low-level and high-level features, improving results compared to state-of-the-art techniques.

1. Introduction

A brain tumor is a development that develops in the tissues surrounding the brain or the skull and has a major impact on human life. A benign or malignant growth can be characterized in two ways. Primary cancers begin within the brain, whereas secondary tumors, known as brain metastasis tumors, are most commonly generated from tumors outside the brain. The most prevalent primary brain tumors include glioma, meningioma, and pituitary adenomas. Meningioma is a slow-growing tumor that develops from the brain, spinal cord membrane layers, and brain. Pituitary adenomas are tumors that emerge in the pituitary gland [1]. These tumors grow unevenly in the brain, putting pressure on the existing tissue. Pressure causes a slew of brain issues that have a negative impact on the body [2]. In contrast to benign tumors, malignant tumors develop unevenly, causing harm to the surrounding tissues. Several approaches can be used to treat brain tumors. In the treatment of brain tumors, surgical methods are commonly used [3]. MRI is chosen over other imaging techniques including computed tomography (CT), positron emission tomography (PET), and X-rays because of its non-interfering characteristic [4,5]. By 2022, 72,

360 people aged 40 and up in the United States are predicted to be identified with a primary brain tumor. A total of 700,000 Americans are thought to be affected by primary brain tumors, with benign tumors accounting for 71% of all instances and malignant tumors accounting for 29% [6]. Tumors are discovered by conventional medical methods such as MRI, and biopsies are taken for examination. MRI is the first medical procedure to detect cancer [7,8]. Fig. 1 depicts two MRI scans of two different brains.

Manually interpreting these photos has become time-consuming, chaotic, and usually wrong as the number of patients has increased. To alleviate this constraint, a computer-aided diagnostic method that limits the cost of brain MRI identification must be established. Several efforts have been made to develop a very efficient and reliable approach for automatically categorizing brain tumors. Handmade qualities are used in traditional machine learning methods, limiting the solution's durability and raising the cost. However, supervised learning models can sometimes outperform unsupervised learning techniques, resulting in an overfitted model that is unsuitable for another huge database. These issues highlight the importance of developing an entirely automated brain tumor categorization based on machine learning. CNN's

* Corresponding author.

E-mail address: saiful05eee@cuet.ac.bd (M.S. Islam).

<https://doi.org/10.1016/j.measen.2023.100694>

Received 22 October 2022; Received in revised form 18 December 2022; Accepted 2 February 2023

Available online 6 February 2023

2665-9174/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

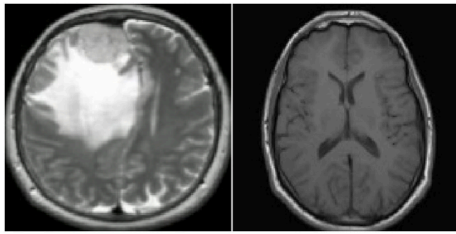


Fig. 1. Two separate brains are scanned using MRI. Healthy on the right and a tumor on the left [9].

design is based on a deep learning model, a neural network that is particularly good at picture recognition and classification [10,11]. The goal of this study is to create a fully self-contained PDCNN model for brain tumor categorization using publicly available Kaggle and Figshare datasets [9,12,13]. This study proposes a brain tumor detection and classification architecture that comprises two simultaneous deep convolutional neural networks. The PDCNN model is quicker over the conditional random field (CRF)-based techniques because convolutions are both precise and efficient procedures in terms of time. Batch normalization is also utilized in the proposed PDCNN framework to normalize the outcomes of preceding layers.

The benefits of parallel pathways are provided by integrating two simultaneous deep convolutional neural networks having two different window sizes, allowing the model to learn global and local features. That's one of the article's contributions. The work's main achievements are displayed in three aspects: 1) A PDCNN has been created as a fully autonomous brain tumor detection and classification architecture by combining two parallel convolutional neural networks. 2) Boosting the identification and categorization performance by including high-level and low-level data, and specific brain characteristics. 3) The dynamical properties provided by the two simultaneous convolutional layers are considerably improved using feature fusion technology.

The remaining of this article will be arranged as follows: Section 2 contains a synopsis of relevant research and a full evaluation of these studies. The suggested PDCNN model is thoroughly detailed in section 3. A complete comparison of the suggested method to current methodologies, and a discussion of the experimental findings, are included in sections 4 and 5. The article closes with section 6, which is the final section of the study.

2. Related work

Depending on the classification of brain tumors, there are a variety of research papers in the literature. Here are some of the works that have been discussed.

Habib [14] has suggested a convolutional neural network to detect brain cancers using the Kaggle binary brain tumor classification dataset-I, used in this article. This approach can achieve an accuracy of 88.7% using a modified neural network architecture [15]. explains the creation of a model that focuses on an artificial CNN for MRI analysis utilizing mathematical formulas and matrix operations. MRI has been used to train this neural network on 155 brain tumors and 98 tumor-free brains. With a 96.7% accuracy rate in the validation data, the model accurately indicates the location of a tumor. Anil et al. have suggested a technique [16] that comprises a classification network that splits the input MRI pictures into two categories: one with a tumor and the other without one. The classifier in this work is retrained using the transfer learning method to identify brain cancers. VGG-19 is the most effective, according to the findings, with an accuracy rate of 95.78%. Muhammad Sajjad et al. have developed a new CNN model to classify brain tumors [17]. Segmentation is performed to pick the tumor spot before using MRI pictures. In the following stage, the dataset is expanded. The recommended CNN is then used to carry out the categorization procedure.

The precision of their classifications is 94.58%. P. Afshar has proposed employing capsule networks as a convolutional neural network structure [18]. Using the Figshare dataset, the suggested scheme focuses on the primary region of tumors and their connections to neighboring tissues, with a precision of 90.89%. In Ref. [19], Mahmoud Khaled Abd-Allah et al. proposed a PDCNNs structure for detecting and classifying gliomas from brain MRI images. The BraTS-2017 dataset is used to test the suggested PDCNNs. The study uses 1200 photos in the training phase of the PDCNN, 150 images in the validation phase of the PDCNN, and 450 photos in the test phase of the PDCNN. The framework has delivered remarkable results in accuracy, sensitivity, and specificity, (97.44%, 97.0%, and 98.0%, respectively).

A CNN is used by A. E. Minarno et al. to detect 3 types of brain tumors on MRI images [20]. In this research, 3264 datasets with detailed photos of glioma tumors (926 photos), meningioma tumors (937 photos), pituitary cancers (901 photos), and others with no tumors (500 photos) are examined. The CNN approach paired with Hyperparameter Tuning is presented to get optimal results in classifying brain tumor varieties. Three different scenarios are used to test the model and in the third model evaluation scenario, the outcome shows an accuracy of 96.00%. Using well-known deep transfer learning models such as DenseNet201, DenseNet169, DenseNet121, MobileNet v2, VGG19, VGG16, and Xception [21], focuses on the construction of a unique brain tumor detection and categorization system. Using a benchmark dataset, the performance of the suggested framework is assessed in terms of accuracy and loss. With DenseNet201, it is noted that training accuracy is 97.49% and validation accuracy is 96.43% and performs better than all other models in terms of accuracy. Using the transfer learning approach in the InceptionV3 and DenseNet201 models, S.U. Habiba et al. explored a classification problem on a public dataset in Ref. [22]. In order to get a good classification outcome and prevent overfitting, data augmentation is used. Six convolutional layers that are densely coupled together to extract features from dense layers make up a deep CNN model. This model's ability to distinguish between the three different forms of brain tumors has a classification accuracy of about 96.3%.

In [23], using machine learning approaches, A. K. Budati et al. describe an automated brain tumor categorization and detection model of brain MRI scans. In order to detect the tumor, the preprocessed pictures are segmented with a C-V model, followed by GLCM feature extraction to extract key features from the segmented images, and classification of the images using the SVM and KNN based on the extracted feature selection. More accurately than the KNN classifier, the SVM classifier performs at a level exceeding 98%. A novel automated system for the detection and classification of brain tumors that are broken down into the following categories: MRI image preprocessing, image segmentation, extraction of features, and image categorization has been proposed in Ref. [24]. Utilizing an adaptive filter, the MRI image noise is removed. The modified K-means clustering algorithm is used to segment the images, and the gray-level co-occurrence matrix is employed for feature extraction. Finally, a deep learning model recurrent convolutional neural network is utilized to categorize glioma, meningioma, non-tumors, and pituitary tumors with 95.17% accuracy.

3. Proposed methodology

The biggest issue before starting treatment is detecting and classifying tumors from brain MRI scans. Although most research on brain tumor diagnosis has concentrated on tumor slicing and positioning techniques, there are few studies on tumor diagnosis as a time-saving technique. Most researchers are using a convolutional neural network to classify brain tumors but it collects features randomly without knowing the global and local features. Since the current approach is inconvenient, time-consuming, and human error-prone, these inconveniences display how obligate it is to execute a fully automated deep learning-based brain tumor classification approach. For brain tumor identification and categorization employing MRI pictures, a

parallel architecture of two CNNs is presented. Fig. 2 shows a block diagram of the PDCNN design. The following is the sequence of events that take place within the suggested structure: the input layer of the PDCNNs receives brain MRI pictures, which get preprocessed to reduce computing complexity. For training purposes, the input images are transformed to 32×32 pixels from a variety of pixel heights and widths. These input photos are converted to grayscale, which helps to reduce complexity. After that, data augmentation is used to create new photos from old ones. To train the proposed network, the data set is separated into training and validation. The PDCNN structure is then used to classify the input pictures, which include local, global, merging, and output routes. At the output path, to carry out the brain tumor categorization process, the softmax function is employed.

3.1. MRI dataset

Three different public dataset of brain MRI pictures are used in this research. For simplicity, the Kaggle platform has been used to obtain the first publicly attainable dataset of binary-class brain MRI pictures; in this paper, this data is referred to as dataset-I [9]. There are 253 brain MRI pictures in this collection, including 98 tumors and 155 non-tumor groups. This study makes use of the Figshare dataset of 233 patients' brain MRI pictures [12]. These brain MRI pictures are taken at two Chinese hospitals (Nanfang Hospital and General Hospital). It contains 3064 MRI scans of the brain (1426 glioma tumors, 708 meningioma tumors, and 930 pituitary tumors); this dataset is identified as dataset-II. Finally, the Kaggle website is also used to obtain the other dataset used in this research [13]; it includes 826, 822, 395, and 827 brain MRI pictures, respectively, of glioma tumor, meningioma tumor, no tumor, and pituitary tumor. In this study, this data is referred to as dataset-III. Table 1 displays the different types of brain MRI pictures found across all datasets.

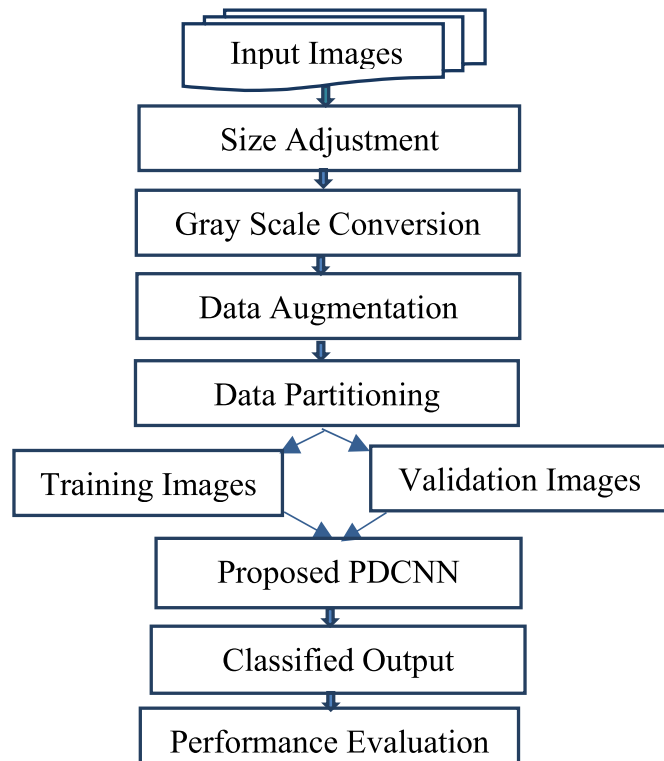


Fig. 2. The framework of the proposed approach.

3.2. Preprocessing data

Data preprocessing is a technique for cleaning data and preparing it for use in a machine-learning model, improving the model's accuracy and efficiency. All the brain images in the MRI datasets do not have the same width, height, or size; to provide uniformity for training reasons, all of the photos are scaled to 32×32 pixels. These input photos are converted to grayscale, which helps to reduce complexity.

3.3. Data augmentation

At this stage, data augmentation is used to expand the amount of data available by modifying the original image because deep learning requires a large amount of data to learn from. It is possible to improve the effectiveness of categorized outcomes by supplementing them. Rotation, scaling, translation, and filtering are all operations that can be applied to images. The filtering procedure is used as a supplement in this article.

The inappropriate information consists of noise being present in all MRI brain pictures, resulting in a poor recognition rate. To provide valuable information, it is required to reduce the noise and unwanted areas. High-frequency noise is seen in MRI pictures, which is generally minimized using a filtering process. The anisotropic diffusion filter is a technique for autonomous noise suppression while maintaining image edges. This filter can be used to remove noise from digital photos while avoiding edge blurring. Table 2 displays the performance of MRI brain images after using an anisotropic diffusion filter.

3.4. Developed Parallel deep convolutional neural networks design

Extraction of features and categorization are the two components of a standard CNN. The CNN models include input, convolutional, pooling, fully connected, and classification layers in their construction, shown in Fig. 3. For classification, fully connected layers and categorization layers are utilized, whereas the characteristics are obtained using convolutional and pooling layers.

CNN has been widely used in image/video recognition and classification in recent years. CNN can extract image features that are either global or local, or both, automatically from input images.

This research proposes a new network topology for brain tumor detection and characterization that consists of two deep convolutional neural networks operating simultaneously which are shown in Fig. 4. The PDCNN structure is used to classify the input pictures, which include local, global, merging, and output routes. To acquire local and global features, the local and global pathways are used, respectively. In the local path, the convolutional layers utilize the modest window size of 5×5 pixels to offer low-level information with in the pictures. The convolutional layers of the global pathway, on the other hand, have a massive number of filters of 12×12 pixels. For each path that down-samples the convolutional layer output, the max-pooling layer is utilized after every convolutional layer. The two pathways are joined by a fusion layer, which creates a single path along with a cascaded link until it entered the ultimate goal. A batch normalization layer precedes a ReLU layer, which is followed by two fully linked layers that are coupled to a dropout layer in the merging route. To deal with the over-fitting problem, a dropout of 0.3 is applied in the initial layers of the model, which decreases as the network grows deeper. At the output path, to carry out the brain tumor categorization process, the softmax function is employed.

In the PDCNN framework, both global and local features obtained from the two parallel stages are incorporated. Dropout is a regularization strategy for preventing overfitting in training data.

The layers used in the proposed PDCNN framework are discussed below-

A. Convolutional Layers The convolutional layer is the most significant component of the recommended PDCNN framework. Aside from

Table 1

Categorization of MRI brain pictures.

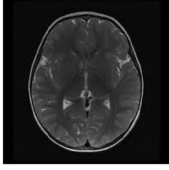
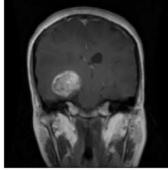
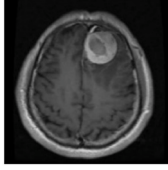
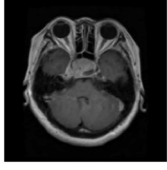
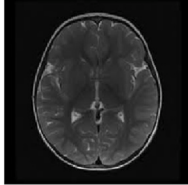
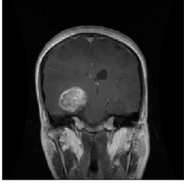
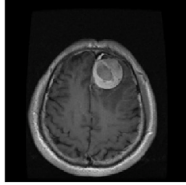
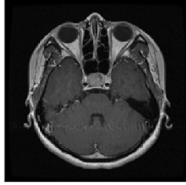
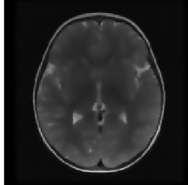
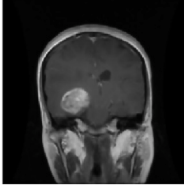
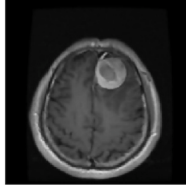
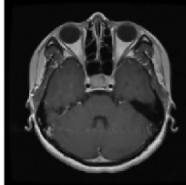
	No Tumor	Glioma Tumor	Meningioma Tumor	Pituitary Tumor
MRI Brain Pictures [13]				

Table 2

Performance of MRI brain images after using an anisotropic diffusion filter.

	No Tumor	Glioma Tumor	Meningioma Tumor	Pituitary Tumor
MRI Brain Pictures [13]				
Anisotropic Diffusion Filtered Pictures				

**Fig. 3.** The normal CNN model.

the initial layer, the convolutional layer uses maps from the previous layer. It generates feature maps as a result.

- B. Max Pooling Layers** The input is down-sampled via max-pooling by reducing its dimension. The number of learned parameters should be reduced as a result of this procedure, lowering the computing cost and helping avoid overfitting.
- C. ReLU Layers** In the hidden layers, ReLU layers are commonly used. If the inputs are greater or exceed 0, the outcomes (x, y) are similar; alternatively, the outcomes are equivalent to 0, as in equation (1).

$$f_y = \max(x, 0) \quad (1)$$

D. Integrating Path The fusion procedure's purpose is to integrate features from all channels and encrypt inter-group knowledge into a reduced feature map. Batch normalization and subsequently ReLU activation follow a merging procedure. The resulting feature map of an integrating method is determined by equation (2).

$$Z = \sigma(BN(f(X))) \quad (2)$$

Where σ refers the ReLU activation function, BN defines the batch normalization function, and the merged extracted features from each layer in the prior routes are represented by $f(X)$.

E. Regularization As shown in equation (3), to reduce variations and regulate the transferred data to the kernel function, batch normalization is performed. PDCNN training is aided by BN, which facilitates weight initialization by providing more feasible activation functions, simplifying the optimization, achieving optimal results faster, and generating adjustable gradients.

$$\hat{m} = \frac{m - E[m]}{\sqrt{\text{Var}[m]}} \quad (3)$$

where $\hat{m}$ denotes normalized activations, m denotes the layer input, E [m] represents the expectation, and Var[m] is the unbiased variance estimate.

Another regularization strategy is a dropout, which ignores neurons that are picked at random and ceases updating their weights during training [19].

Details on the PDCNN model that has been proposed, are represented in Table 3.

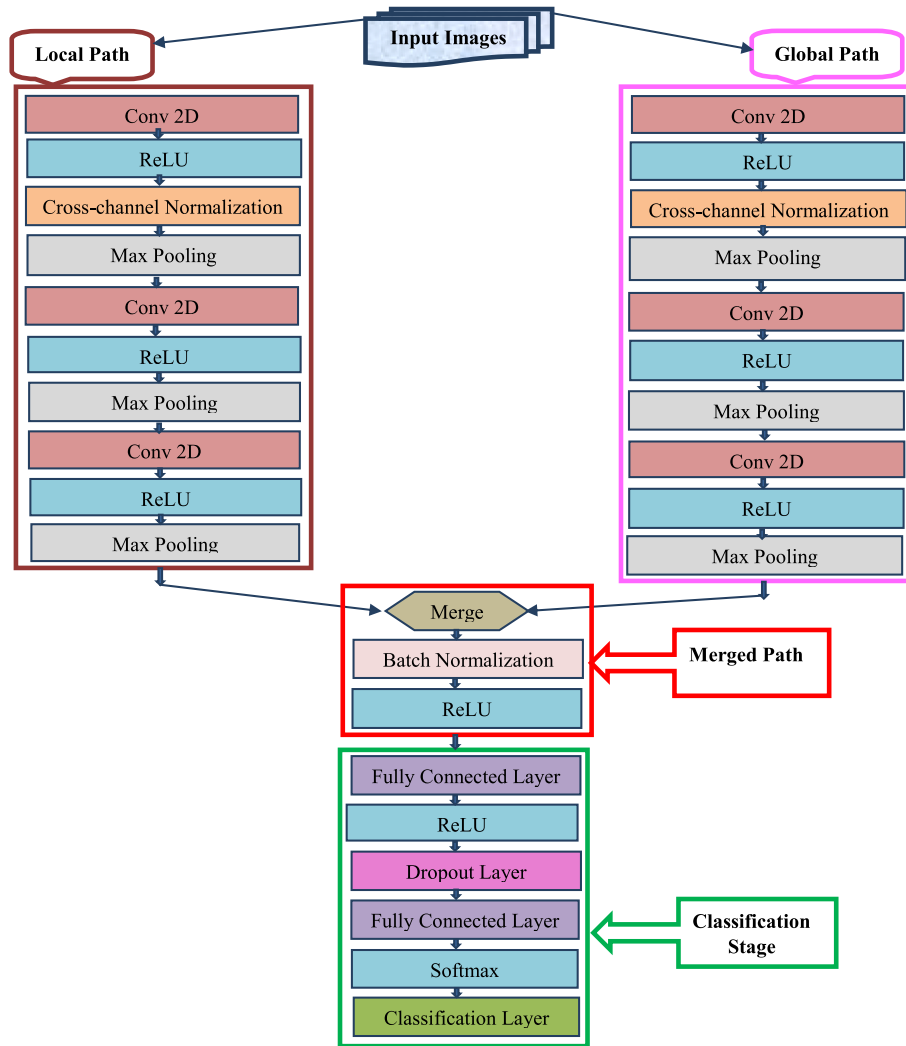


Fig. 4. The PDCNN model contains four stages in its overall structure: local, global, merging, and output stages in order to categorize the input image.

4. Implementation and evaluation

The PDCNN's implementation code is executed in MATLAB. The computer has a 3.2 GHz Intel Core i5 processor, 8 GB of RAM, and a Windows operating system installed.

4.1. Classification results

The proposed two simultaneous deep CNN architecture for tumor identification and categorization is validated using three datasets: dataset-I [9], dataset-II [12], and dataset-III [13]. As shown in Figs. 5–7, multiple ratios are provided for training and testing of the proposed design, including (90:10), (80:20), (70:30), (60:40), and (50:50).

The accuracy of brain tumor classification using the PDCNN model for multiple ratios is shown in Fig. 5 for binary dataset-I. The number of epochs, iterations, elapsed time, and accuracy is calculated for each ratio. At a 90:10 training-testing ratio, the maximum accuracy achieved is 96.00%. This result is obtained after 90 epochs and 90 iterations, with a total execution time of 361 s. A change in the training-testing ratio causes a change in accuracy. The accuracy of the PDCNN model improves after applying augmentation to the binary classification dataset-I. When the training-testing ratio is 90:10, the PDCNN model's accuracy climbs to 97.33% with data augmentation from 96%. The bold numbers represent the greatest possible outcome.

The performance of brain tumor categorization using the PDCNN

model is shown in Fig. 6 for multi-class Figshare dataset- II for many ratios. The number of epochs, iterations, elapsed time, and accuracy is calculated for each ratio. At a 90:10 training-testing ratio, the maximum accuracy achieved is 96.10%. This result is achieved after 55 epochs and 1155 iterations, with a total execution time of 4169 s. A change in the training-testing ratio causes a change in accuracy. The accuracy of the PDCNN model improves after adding augmentation to the multi-class Figshare dataset- II. The PDCNN model's accuracy rises to 97.60% with data augmentation when the training-testing ratio is 90:10. The bold numbers represent the greatest possible outcome.

Fig. 7 shows the efficiency of brain tumor classification using the PDCNN model over several ratios for the multi-class Kaggle dataset- III. The number of epochs, iterations, elapsed time, and accuracy is calculated for each ratio. At an 80:20 training-testing ratio, the greatest accuracy achieved is 95.60%. This result is obtained after 50 epochs and 800 iterations, with a total execution time of 3465 s. A change in the training-testing ratio causes a change in accuracy. The accuracy of the PDCNN model improves after adding augmentation to the multi-class Kaggle dataset- III. The PDCNN model's accuracy rises to 98.12 percent from 95.60% utilizing data augmentation when the training-testing ratio is 80:20. The bold numbers represent the greatest possible outcome.

Table 3
Details on the PDCNN model that has been proposed.

Layer No.	Layer Type	Properties	Activations	Learnable	Total Learnable
1	Image Input	$32 \times 32 \times 1$ images with 'zero center' normalization	$32 \times 32 \times 1$	–	0
2	Convolutional	$128 \ 5 \times 5$ convolutions with stride [2,2] and padding 'same'	$16 \times 16 \times 128$	Weights: $5 \times 5 \times 1 \times 128$ Bias: $1 \times 1 \times 128$	3328
3	ReLU	ReLU	$16 \times 16 \times 128$	–	0
4	Cross Channel Normalization	Cross-channel normalization with 5 channels per element	$16 \times 16 \times 128$	–	0
5	Max	2×2 max pooling with stride [2,2] and padding 'same'	$8 \times 8 \times 128$	–	0
6	Convolutional	$96 \ 5 \times 5$ convolutions with stride [2,2] and padding 'same'	$4 \times 4 \times 96$	Weights: $5 \times 5 \times 128 \times 96$ Bias: $1 \times 1 \times 96$	307,296
7	Convolutional	$128 \ 12 \times 12$ convolutions with stride [2,2] and padding 'same'	$16 \times 16 \times 128$	Weights: $12 \times 12 \times 1 \times 128$ Bias: $1 \times 1 \times 128$	18,560
8	ReLU	ReLU	$4 \times 4 \times 96$	–	0
9	Max	2×2 max pooling with stride [2,2] and stride 'same'	$2 \times 2 \times 96$	–	0
10	Convolutional	$96 \ 5 \times 5$ convolutions with stride [2,2] and padding 'same'	$1 \times 1 \times 96$	Weights: $5 \times 5 \times 96 \times 96$ Bias: $1 \times 1 \times 96$	230,496
11	ReLU	ReLU	$1 \times 1 \times 96$	–	0
12	Max	2×2 max pooling with stride [2,2] and stride 'same'	$1 \times 1 \times 96$	–	0
13	ReLU	ReLU	$16 \times 16 \times 128$	–	0
14	Cross Channel Normalization	cross channel normalization with 5 channels per element	$16 \times 16 \times 128$	–	0
15	Max	2×2 max pooling with stride [2,2] and stride 'same'	$8 \times 8 \times 128$	–	0
16	Convolutional	$96 \ 12 \times 12$ convolutions with stride [2,2] and padding 'same'	$4 \times 4 \times 96$	Weights: $12 \times 12 \times 128 \times 96$ Bias: $1 \times 1 \times 96$	1,769,568
17	ReLU	ReLU	$4 \times 4 \times 96$	–	0
18	Max	2×2 max pooling with stride [2,2] and stride 'same'	$2 \times 2 \times 96$	–	0
19	Convolutional	$96 \ 12 \times 12$ convolutions with stride [2,2] and padding 'same'	$1 \times 1 \times 96$	Weights: $12 \times 12 \times 96 \times 96$ Bias: $1 \times 1 \times 96$	1,327,200
20	ReLU	ReLU	$1 \times 1 \times 96$	–	0
21	Max	2×2 max pooling with stride [2,2] and stride 'same'	$1 \times 1 \times 96$	–	0
22	Addition	Elements-wise addition of 2 inputs	$1 \times 1 \times 96$	–	0
23	Batch Normalization	Batch Normalization	$1 \times 1 \times 96$	offset: $1 \times 1 \times 96$ Scale: $1 \times 1 \times 96$	192
24	ReLU	ReLU	$1 \times 1 \times 96$	–	0
25	Fully	512 fully connected layer	$1 \times 1 \times 512$	Weights: 512×96 Bias: 512×1	49,664
26	ReLU	ReLU	$1 \times 1 \times 512$	–	0
27	Dropout	30% dropout	$1 \times 1 \times 512$	–	0
28	Fully	2 fully connected layer	$1 \times 1 \times 2$	Weights: 2×512 Bias: 2×1	1026
29	Softmax	Softmax	$1 \times 1 \times 2$	–	0
30	Classification Output	–	$1 \times 1 \times 2$	–	0

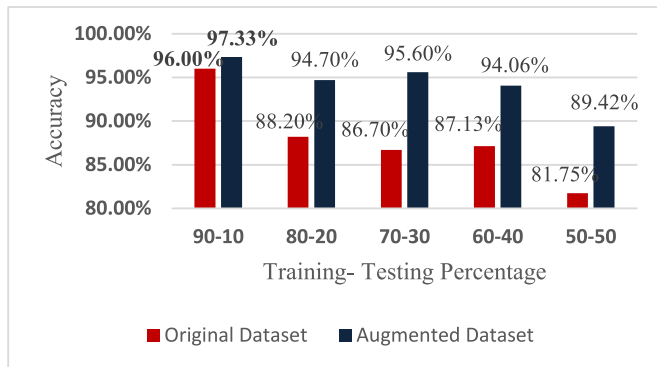


Fig. 5. Visualization of PDCNN accuracy on multiple training and testing ratios for binary brain tumor classification dataset-I.

4.2. Performance evaluation

The outcomes of a categorization system are described using a confusion matrix. The following factor is used to assess performance [25–27].

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{(\text{Positive} + \text{Negative})}$$

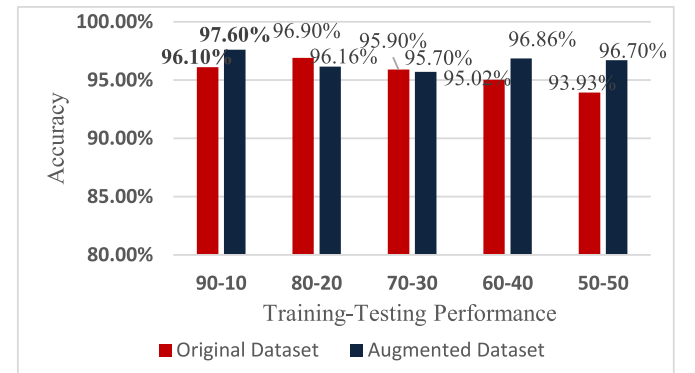


Fig. 6. Visualization of PDCNN accuracy on multiple training and testing ratios for multi-class Figshare dataset- II.

$$\text{Precision} = \frac{\text{True Positive}}{(\text{True Positive} + \text{False Positive})}$$

$$\text{Recall} = \frac{\text{True Negative}}{(\text{False Positive} + \text{True Negative})}$$

$$\text{F1 Score} = \frac{2 * \text{True Positive}}{(2 * \text{True Positive} + \text{False Positive} + \text{False Negative})}$$

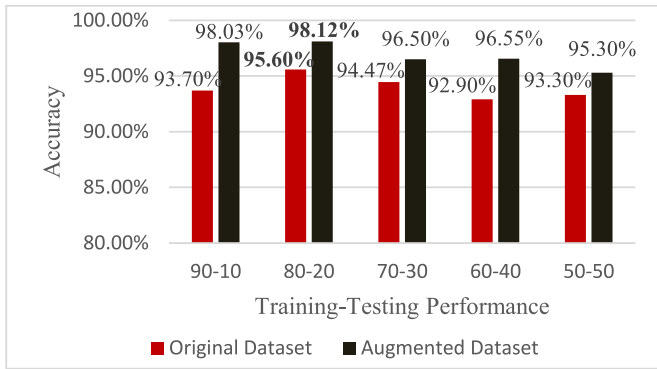


Fig. 7. Visualization of PDCNN accuracy on multiple training and testing ratios for multi-class Kaggle dataset-III.

Fig. 8 illustrates that the PDCNN model outperforms the standard CNN model in terms of accuracy, precision, recall, and F1 score. If augmentation is done to binary classification dataset-I to boost the efficiency of the suggested PDCNN model, the values of the performance indicators will grow even more.

Fig. 9 illustrates that the PDCNN model outperforms the standard CNN model in terms of accuracy, precision, recall, and F1 score. If augmentation is done to multi-class Figshare dataset-II to boost the efficiency of the suggested PDCNN model, the values of the performance indicators will grow even more.

Fig. 10 illustrates that the PDCNN model outperforms the standard CNN model in terms of accuracy, precision, recall, and F1 score. If augmentation is done to multi-class Kaggle dataset-III to boost the efficiency of the suggested PDCNN model, the values of the performance indicators will grow even more.

4.3. Error rate calculation

The error rate is calculated as a percentage of each incorrect prediction divided by the total number of data sets. Accuracy and error rate are complementary. For example, Accuracy = 1- Error Rate and Error Rate = 1- Accuracy.

Fig. 11 shows the CNN model with the greatest error rate for the original dataset. The PDCNN model is used to increase the performance of the CNN model. Compared to the CNN model, the suggested PDCNN model has a lower error rate. When augmentation is applied to the PDCNN model's input dataset, the error rate is lowered even more and the model's performance is significantly improved.

Fig. 12 shows the CNN model with the greatest error rate for the original dataset. The PDCNN model is used to increase the performance of the CNN model. Compared to the CNN model, the suggested PDCNN model has a lower error rate. When augmentation is applied to the

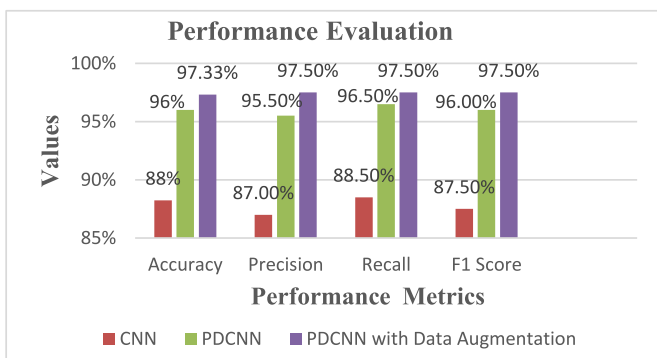


Fig. 8. Performance parameters comparison of accuracy, precision, recall and F1-score with individual models using binary classification dataset-I.

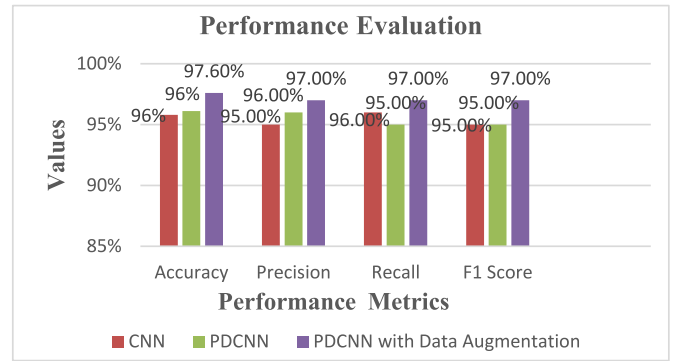


Fig. 9. Performance parameters comparison of accuracy, precision, recall and F1-score with individual models using Figshare dataset- II containing 3 types of brain tumor images.

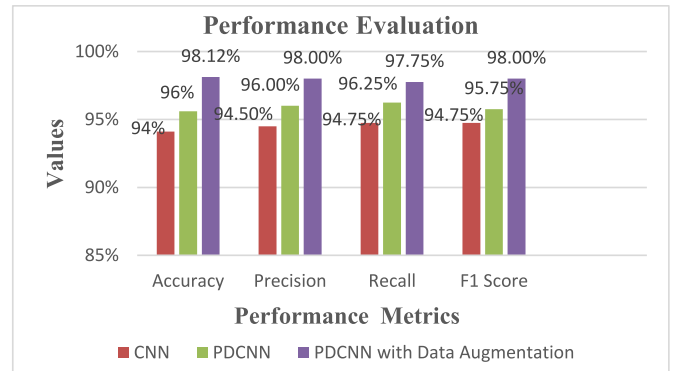


Fig. 10. Performance parameters comparison of accuracy, precision, recall and F1-score with individual models using Kaggle dataset- III containing 4 types of brain tumor images.

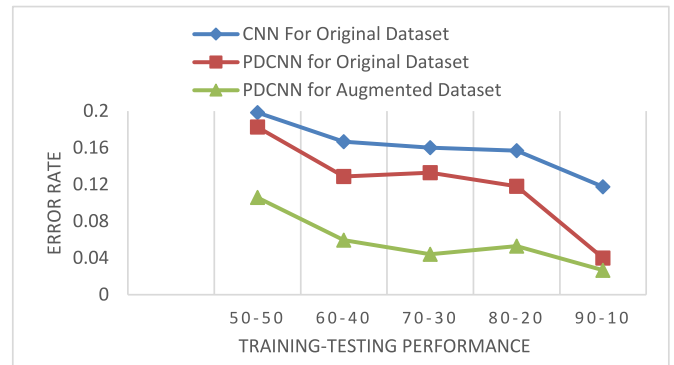


Fig. 11. Training-testing performance vs error rate for binary brain tumor classification dataset-I.

PDCNN model's input dataset, the error rate is lowered even more and the model's performance is significantly improved.

Fig. 13 shows the CNN model with the greatest error rate for the original dataset. The PDCNN model is used to increase the performance of the CNN model. Compared to the CNN model, the suggested PDCNN model has a lower error rate. When augmentation is applied to the PDCNN model's input dataset, the error rate is lowered even more and the model's performance is significantly improved.

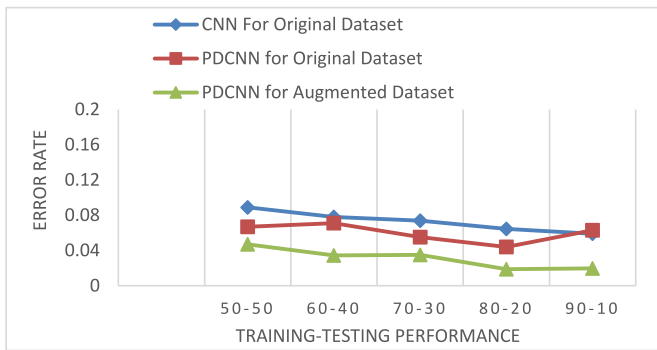


Fig. 12. Training-testing performance vs error rate for multi-class Figshare dataset- II.

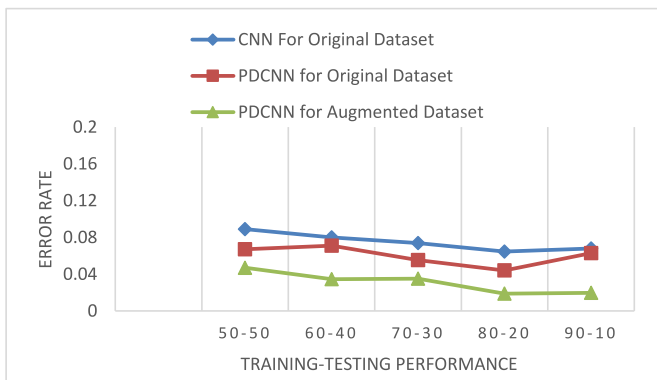


Fig. 13. Training-testing performance vs error rate for multi-class Kaggle dataset- III.

4.4. Result analysis

The suggested PDCNN approach is presented in full in this section.

Table 4 displays the accuracy, error, time, and kappa values for the CNN and PDCNN models using the original dataset-I and enhanced dataset-I, respectively. The kappa statistic is a measure of how well the instances classified by the classification model matched the data designated as ground truth after accounting for the arbitrary classifier's expected accuracy. The PDCNN model's kappa is higher than the CNN model's kappa. After applying augmentation to the image data of both the CNN and PDCNN models, the error rate has decreased. However, after applying augmentation to the input pictures of both the CNN and PDCNN models, the elapsed time has increased.

Fig. 14 shows that the suggested PDCNN model's accuracy improves after augmentation when compared to the original photos. When data augmentation is included, the proposed model's accuracy rises to 97.30% from 96.00%.

Table 5 displays the accuracy, error, time, and kappa values for the CNN and PDCNN models using the original dataset-II and enhanced dataset-II, respectively. The kappa statistic is a measure of how well the

Table 4

Suggested tumor type categorization outcomes using binary classification dataset-I.

Method	Original Dataset I	Augmented Dataset I	Measures			
			Accuracy (%)	Error (%)	Time (s)	Kappa
CNN	✓		88.24	11.76	37	0.749
		✓	97.33	2.67	121	0.944
PDCNN	✓		96.00	4.00	361	0.918
		✓	97.33	2.67	861	0.944

instances classified by the classification model matched the data designated as ground truth after accounting for the arbitrary classifier's expected accuracy. The PDCNN model's kappa is higher than the CNN model's kappa. After applying augmentation to the image data of both the CNN and PDCNN models, the error rate has decreased. However, after applying augmentation to the input pictures of both the CNN and PDCNN models, the elapsed time has increased.

Fig. 15 shows that the suggested PDCNN model's accuracy improves after augmentation when compared to the original photos. When data augmentation is included, the proposed model's accuracy rises to 97.60% from 96.10%.

Table 6 displays the accuracy, error, time, and kappa values for the CNN and PDCNN models using the original dataset-III and enhanced dataset-III, respectively. The PDCNN model's kappa is higher than the CNN model's kappa. After applying augmentation to the image data of both the CNN and PDCNN models, the error rate has decreased. However, after applying augmentation to the input pictures of both the CNN and PDCNN models, the elapsed time has increased.

Fig. 16 shows that the suggested PDCNN model's accuracy improves after augmentation when compared to the original photos. When data augmentation is included, the proposed model's accuracy rises to 98.10% from 95.60%.

4.5. Comparison of the suggested model to previous efforts on the subject

At the conclusion of the suggested method's validation, a detailed comparison is made. Table 7 represents a quick comparison.

5. Discussion

Manually evaluating MRI pictures has become time-consuming, chaotic, and usually erroneous as the number of patients has increased. Handmade attributes are used in traditional machine learning methods, limiting the solution's endurance and increasing the cost. However, supervised learning models can sometimes outperform unsupervised learning techniques, resulting in an overfitted model that is unsuitable for another huge database. The importance of developing an entirely automated brain tumor categorization based on machine learning is highlighted by these issues. This research offers a new method for brain tumor identification and categorization that consists of two simultaneous deep convolutional neural networks. Local, global, merging, and output pathways are all part of the PDCNN framework. Grayscale conversion is applied to the input photos, which helps to simplify the process. Then, using data augmentation, new photos are created from existing ones. The convolutional layers in the local path give local information to the images by using a modest window size of 5×5 pixels. The convolutional layers of the global path, on the other hand, use a massive window size of 12×12 pixels. For each path that downsamples the convolutional layer output, the max-pooling layer is utilized after every convolutional layer. The two pathways are joined by a fusion layer, which creates a single path along with a cascaded link until it entered the ultimate goal. A batch normalization layer precedes a ReLU layer, which is followed by two fully linked layers that are coupled to a dropout layer in the merging route. At the output path, to carry out the brain tumor categorization process, the softmax function is employed. Dropout is a regularization technique that is also used to prevent overfitting of the training data.

Using three datasets, the technique delivers very accurate brain tumor categorization. Each dataset's accuracy, precision, recall, and F1 score are determined. The performance of brain tumor classification using the PDCNN model for multiple ratios is shown in Figs. 5–7 for three types of datasets. For datasets I and II, the maximum accuracy is achieved when the training-testing ratio is 90:10.; however, for dataset-III, the best accuracy is achieved when the training-testing ratio is 80:20. A change in the training-testing ratio causes a change in accuracy. The accuracy of the PDCNN model improves after applying augmentation to

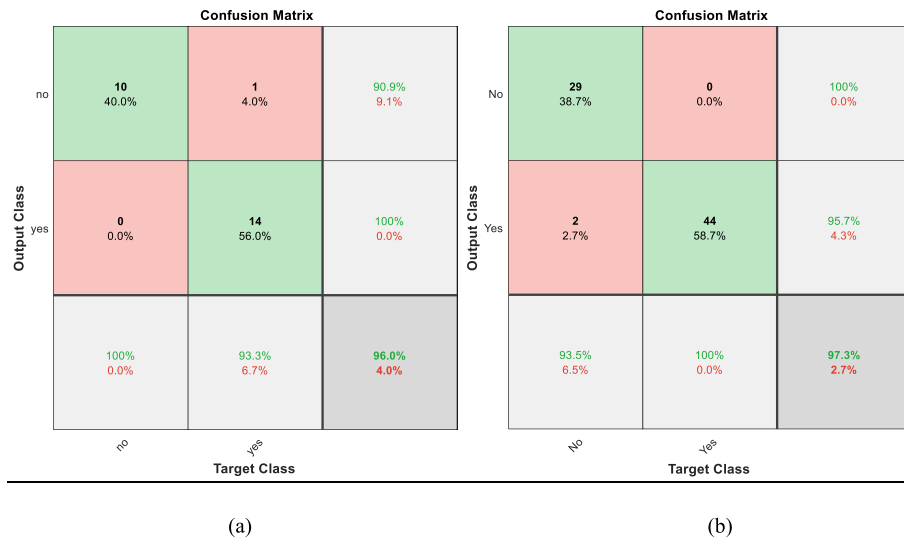


Fig. 14. Confusion matrix of PDCNN using binary classification dataset-I (a) confusion matrix of original dataset-based classification (b) confusion matrix of the original dataset with augmentation-based classification.

Table 5

Suggested tumor type categorization outcomes using multiclass Figshare dataset-II.

Method	Original Dataset II	Augmented Dataset II	Measures			
			Accuracy (%)	Error (%)	Time (s)	Kappa
CNN	✓		95.80	4.20	833	0.933
		✓	97.90	2.10	1627	0.967
PDCNN	✓		96.10	3.90	4169	0.938
		✓	97.60	2.40	5803	0.962

all three types of datasets. The bold numbers represent the greatest possible outcome.

Figs. 8–10 indicate that the PDCNN model outperforms the standard CNN model in terms of accuracy, precision, recall, and F1 score. When augmentation is used on three different types of datasets to improve the performance of the proposed PDCNN model, the performance metrics improve even more.

The original dataset's CNN model has the maximum error rate, as

seen in Figs. 11–13. The PDCNN model is used to increase the performance of the CNN model. Compared to the CNN model, the suggested PDCNN model has a lower error rate. When augmentation is applied to the PDCNN model's input dataset, the error rate is lowered even more and the model's performance is improved.

The results in Figs. 14–16 show that augmentation improves the accuracy of the proposed PDCNN model when compared to the original images. At the conclusion of the suggested method's validation, a

Table 6

Suggested tumor type categorization outcomes using multiclass Kaggle dataset-III.

Method	Original Dataset III	Augmented Dataset III	Measures			
			Accuracy (%)	Error (%)	Time (s)	Kappa
CNN	✓		94.10	5.90	629	0.919
		✓	97.70	2.30	783	0.968
PDCNN	✓		95.60	4.40	3465	0.940
		✓	98.12	1.88	6430	0.974

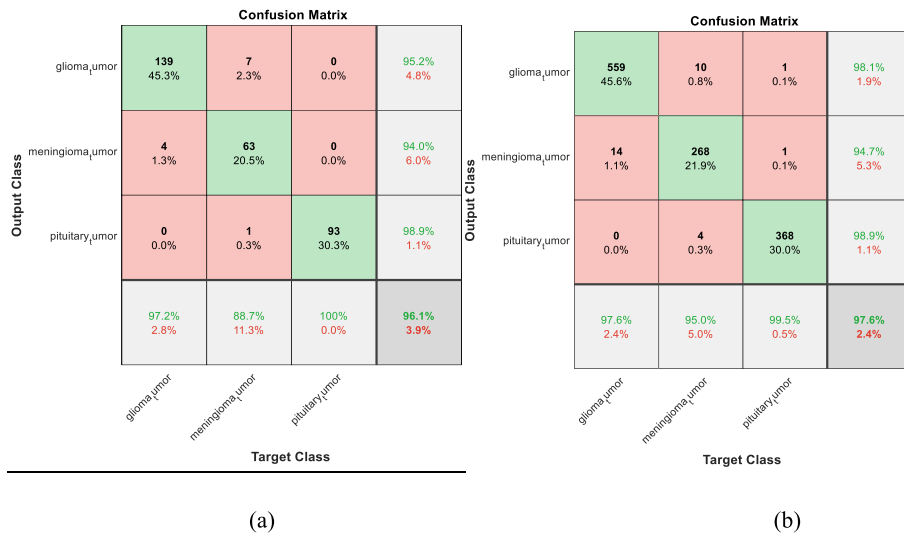


Fig. 15. Confusion matrix of PDCNN using Figshare dataset-II (a) confusion matrix of the original dataset-based classification (b) confusion matrix of the original dataset with augmentation-based classification.

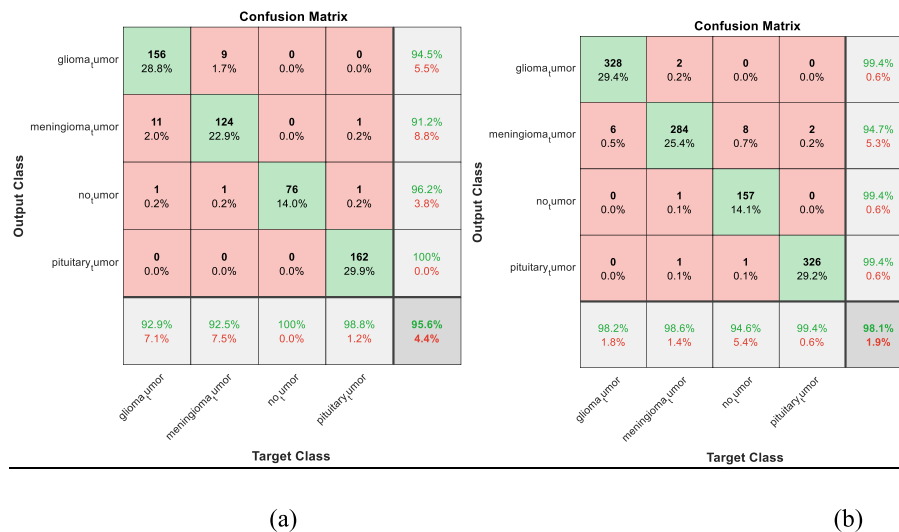


Fig. 16. Confusion matrix of PDCNN using Kaggle dataset-III (a) confusion matrix of the original dataset-based classification (b) confusion matrix of the original dataset with augmentation-based classification.

Table 7
Comparison with existing techniques those used Kaggle and Figshare datasets.

No	Authors	Methodology	Year	Dataset	Accuracy
1.	P. Afshar et al. [18]	Capsule Networks	2019	Figshare Dataset-II	90.89%
2.	Chirodip Lodh Choudhury et al. [26]	CNN	2020	Binary Dataset-I	96.08%
3.	H. H. Sultan et al. [25]	Resize + Augmentation + CNN + Hyperparameter Tuning	2019	Figshare Dataset-II	96.13%
4.	Suhib et al. [15]	Gray + Resize + Flatten + CNN	2020	Binary Dataset-I	96.7%
5.	Agus Eko Minarno et al. [20]	Resize + Augmentation + CNN + Hyperparameter Tuning	2021	Kaggle Dataset-III	96.00%
6.	Priyansh et al. [29]	CNN Based Transfer Learning Approach	2021	Binary Dataset-I	Resnet-50-95%, VGG-16-90%, Inception-V3-55%
7.	T.Rahman et al. [30]	Resize + Gray + Augmentation + Binary + CNN	2022	Binary Dataset-I	96.9%
8.	Proposed Structure	Resize + Gray scale conversion + Augmentation + PDCNN	-	Binary Dataset-I	97.33%
				Figshare Dataset-II	97.60%
				Kaggle Dataset-III	98.12%

detailed comparison is carried out. The results show that the proposed simultaneous network topology exceeds previously published detection and classification methodologies.

6. Conclusion and future work

Tumors are a terrible menace to humans because cancerous cells can attack nearby tissue and spread to other parts of the body. The significance of early brain tumor identification in providing appropriate medical treatment has long been recognized. This research introduced a

PDCNN framework for identifying and categorizing brain tumors from MRI pictures. On the Kaggle dataset, the suggested PDCNN is tested for binary classification and multi-class classification, with four types of tumor pictures in the Kaggle dataset and three types of tumor pictures in the Figshare dataset. For three different types of brain tumor datasets, the structure delivered remarkable accuracy, precision, recall, and F1 score. The proposed structure will be expanded in the future to use a 3D structure for tumor identification using 3D brain MRI pictures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The Chittagong University of Engineering and Technology is funding the endeavor under the project CUET/DRE/2020-21/ETE/009.

References

- [1] T.A. Sadoon, M.H.A. Al-Hayani, Deep learning model for glioma, meningioma and pituitary classification, *Int. J. Adv. Appl. Sci.* 10 (1) (March 2021) 88–98.
- [2] W. Grisold, A. Grisold, Cancer around the brain, *Neuro-oncol. pract.* 1 (1) (2014) 13–21.
- [3] M. Kheirollahi, S. Dashti, Z. Khalaj, F. Nazemroaia, P. Mahzouni, Brain tumors: special characters for research and banking, *Adv. Biomed. Res.* (6 January 2015).
- [4] J. Lukoff, J. Olmos, Minimizing medical radiation exposure by incorporating a new radiation “vital sign” into the electronic medical record: quality of care and patient safety, *Perm. J.* 27 (September) (2017).
- [5] R. Vankdothu, M.A. Hameed, Brain tumor MRI images identification and classification based on the recurrent convolutional neural network, *Measurement: Sensors* 24 (2022) 28. September.
- [6] [Online]. Available: Quick Brain Tumor Facts |, Quick Brain Tumor Facts”, National Brain Tumor Society, 2022.
- [7] S. Irshaidat, R. Duwairi, Brain tumor detection using artificial convolutional neural networks, in: 2020 11th International Conference on Information and Communication Systems, ICICS), 2020, pp. 197–203.
- [8] M. Toğaçar, B. Ergen, Z. Cömert, BrainMRNet: brain tumor detection using magnetic resonance images with a novel convolutional neural network model, *Med. Hypotheses* 134 (2020) 1. January.
- [9] Brain MRI Images for Brain Tumor Detection | Kaggle.

- [10] R. Yamashita, M. Nishio, R.K.G. Do, K. Togashi, Convolutional Neural Networks: an Overview and Application in Radiology," *Insights into Imaging*, 2018, pp. 611–629.
- [11] Z. A. Sejuti, M. S. Islam, "An Efficient Method to Classify Brain Tumor using CNN and SVM," 2021 International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD), pp. 259-263, 2021.
- [12] Jun Cheng, 2017 Figshare dataset https://figshare.com/articles/brain_tumor_dataset/1512427.
- [13] Brain Tumor MRI Dataset | Kaggle.
- [14] Brain Tumor Detection Using Convolutional Neural Networks | by Mohamed Ali Habib | Medium 25 November, 2021.
- [15] S. Irsheidat, R. Duwairi, Brain tumor detection using artificial convolutional neural networks, in: 2020 11th International Conference on Information and Communication Systems, ICICS), 2020, pp. 197–203.
- [16] A. Anil, A. Raj, H.A. Sarma, N. Chandran R, P.L. Deepa, Brain tumor detection from brain MRI using deep learning, *Int. J. Innovat. Res. Appl. Sci. Eng. (IJIRASE)* 3 (Issue 2) (August 2019) 68–73.
- [17] M. Sajjad, S. Khan, K. Muhammad, W. Wu, A. Ullah, S.W. Baik, Multi-grade brain tumor classification using deep CNN with extensive data augmentation, *J. comput. sci.* 30 (2019) 174–182.
- [18] P. Afshar, K.N. Plataniotis, A. Mohammadi, Capsule networks for brain tumor classification based on MRI images and coarse tumor boundaries, in: 2019 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2019, pp. 1368–1372.
- [19] M.K. Abd-Ellah, A.I. Awad, H.F.A. Hamed, A.A.M. Khalaf, "Parallel deep CNN structure for glioma detection and classification via brain MRI Images.", in: 2019 31st International Conference on Microelectronics, ICM), 2019, pp. 304–307.
- [20] A.E. Minarno, M.H.C. Mandiri, Y. Munarko, Hariyady,"Convolutional neural network with hyperparameter tuning for brain tumor classification, *Kinetik: Game Technol. Inf. Syst. Comput. Network, Comput. Electron. Control* 6 (2) (May 2021) 127–132.
- [21] P. Rajak, A.S. Jangde, G.P. Gupta, Towards Design of Brain Tumor Detection Framework Using Deep Transfer Learning Techniques," *Convergence of Big Data Technologies and Computational Intelligent Techniques*, IGI Global, 2023, pp. 90–103.
- [22] S.U. Habiba, M.K. Islam, L. Nahar, F. Tasnim, M.S. Hossain, K. Andersson, Brain-DeepNet: a deep learning based classifier for brain tumor detection and classification, in: International Conference on Intelligent Computing & Optimization, Springer, 2023, pp. 550–560.
- [23] A.K. Budati, R.B. Katta, An automated brain tumor detection and classification from MRI images using machine learning techniques with IoT, *Environ. Dev. Sustain.* 24 (9) (2022) 10570–10584.
- [24] R. Vankdothu, M.A. Hameed, Brain tumor MRI images identification and classification based on the recurrent convolutional neural network, *Measurement: Sensors* 24 (December 2022).
- [25] H.H. Sultan, N.M. Salim, W. Al-Atabany, Multi-classification of brain tumor images using deep neural network, *IEEE Access* (2019) 69215–69225. May 27.
- [26] D. Lamrani, B. Cherradi, O.E. Gannour, M.A. Bouqentar, L. Bahatti, Brain tumor detection using MRI images and convolutional neural network, *Int. J. Adv. Comput. Sci. Appl.* 13 (No. 7) (2022).
- [27] A. Nayan, A.N. Mozumder, M.R. Haque, F.H. Sifat, K.R. Mahmud, A.K.A. Azad, M. G. Kibria, A deep learning approach for brain tumor detection using magnetic resonance imaging, *Int. J. Electr. Comput. Eng.* 13 (No. 1) (February 2023) 1039–1047.
- [29] S. Priyansh, A. Maheshwari, S. Maheshwari, Predictive modeling of brain tumor: a deep learning approach, *Innovations Comput. Intell. Comput. Vis.* 3 (Issue 2) (2021) 275–285.
- [30] T. Rahman, M.S. Islam, "MRI brain tumor classification using deep convolutional neural network," in: 2022 International Conference On Innovations In Science, Engineering And Technology (ICISSET), Chittagong, Bangladesh, 2022, pp. 451–456.